Polar $\rightarrow$ Rectangular: $x = r\cos\theta; \quad y = r\sin\theta$

Rectangular $\rightarrow$ Polar:

$$x^2 + y^2 = r^2; \quad r = \pm\sqrt{x^2 + y^2}$$

$$\tan\theta = \frac{y}{x}; \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

How to use calculator to figure out θ :

Quadrant II $x < 0;\ y > 0$ $\theta = \tan^{-1}(y/x) + \pi$	Quadrant I $x > 0;\ y > 0$ $\theta = \tan^{-1}(y/x)$
$x < 0;\ y < 0$ $\theta = \tan^{-1}(y/x) + \pi$ Quadrant III	$x > 0;\ y < 0$ $\theta = \tan^{-1}(y/x) + 2\pi$ Quadrant IV

Also:

If (x, y) is on positive $y-$axis ($x = 0$ and $y > 0$): $\theta = \pi/2$.

If (x, y) is on negative $y-$axis ($x = 0$ and $y < 0$): $\theta = 3\pi/2$.

If (x, y) is on positive $x-$axis ($x > 0$ and $y = 0$): $\theta = 0$.

If (x, y) is on negative $x-$axis ($x < 0$ and $y = 0$): $\theta = \pi$.

Polar Equations of Conics:

$$r = \frac{ed}{1 \pm e\cos\theta} \quad \text{or} \quad r = \frac{ed}{1 \pm e\sin\theta}$$

where e $(\text{eccentricy}) > 0$ and

$|d| =$ distance between the focus at the pole and directrix.

Note: Focus is at the pole.

Classification of Conics by Eccentricity

Let $e =$ eccentricity.

a) The conic is a ellipse for $0 < e < 1$.

b) The conic is a parabola for $e = 1$.

c) The conic is a hyperbola for $e > 1$.

Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \dfrac{ed}{1 + e\sin\theta}$

2) Horizontal directrix below the pole: $r = \dfrac{ed}{1 - e\sin\theta}$

3) Vertical directrix to the right of the pole: $r = \dfrac{ed}{1 + e\cos\theta}$

4) Vertical directrix to the left of the pole: $r = \dfrac{ed}{1 - e\cos\theta}$

If $\sin(x) = c$,

then: $x = \sin^{-1}(c)$, $\quad x = \pi - \sin^{-1}(c)$,

$x = \sin^{-1}(c) + 2\pi$, $\quad x = 3\pi - \sin^{-1}(c)$

For $\cos(x) = c$,

then: $x = \cos^{-1}(c)$, $\quad x = 2\pi - \cos^{-1}(c)$,

$x = \cos^{-1}(c) + 2\pi$, $\quad x = 4\pi - \cos^{-1}(c)$

Area of Polar Region:

$$\text{Area} = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, d\theta$$

Arc Length

$$\text{Arc Length} = \int_a^b \sqrt{1 + [\,f'(x)]^2}\,dx$$ for rectangular equation

$$\text{Arc Length} = \int_c^d \sqrt{[dx/dt]^2 + [dy/dt]^2} \cdot dt$$ for parametric equations

$$\text{Arc Length} = \int_\alpha^\beta \sqrt{f[(\theta)]^2 + [f'(\theta)]^2}\,d\theta = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$$ for polar equation